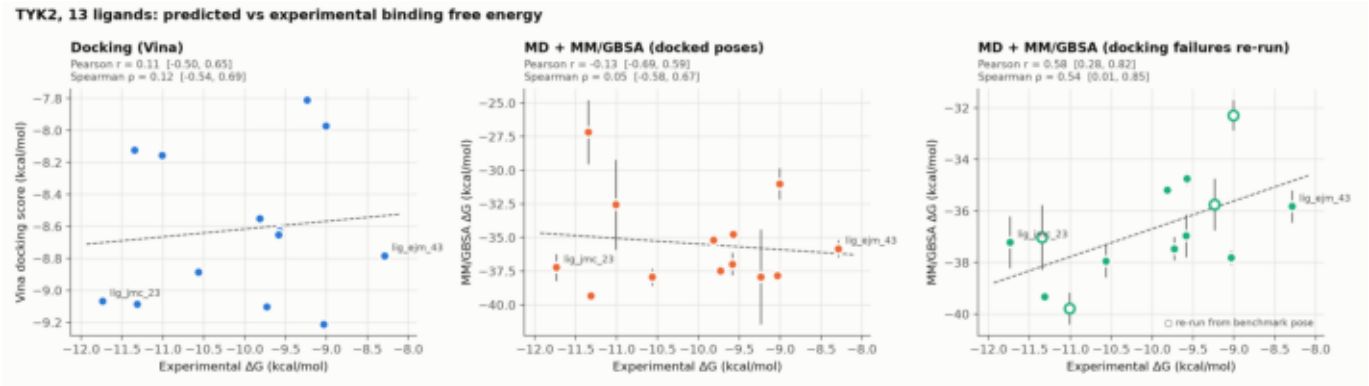
Case study: does MD rescoring beat docking?

TYK2 kinase · 13 inhibitors with measured Ki · open-source pipeline

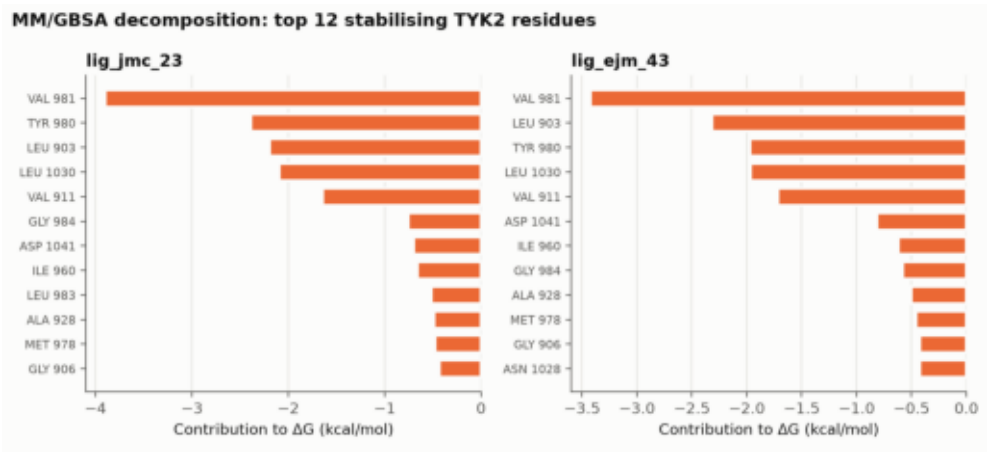
Question

For a congeneric inhibitor series, does MD + MM/GBSA rank compounds better than the docking score?



Results

- Vina placed 9/13 ligands within 2 Å of the benchmark pose.
- MD + MM/GBSA from the docked poses gave Pearson r = -0.13 against experiment, versus r = 0.11 for the docking score (the 95% CIs overlap, so with 13 compounds this is a trend, not a proof).
- Re-running the 4 misdocked ligands from the benchmark pose changed r to 0.58 (+0.71), so wrong docking poses were a large part of the MM/GBSA error.



Method

AutoDock Vina docking → GAFF2/AM1-BCC ligand parameters → GROMACS MD (ff14SB, TIP3P, 0.15 M NaCl; 3 × 10 ns replicas per complex) → MM/GBSA (gmx_MMPBSA, igb=5, mbondi2) with per-residue decomposition. Bootstrap 95% confidence intervals on all correlations.

What you get when I run this for your series

✔ Ranked compounds with error bars, not single numbers
✔ Binding-mode check that flags poses docking got wrong
✔ Per-residue maps of what drives binding
✔ Paper-ready methods, figures and all input files

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